Yang–Mills Structure Identification for the Reconstructed Connection

A Proposed Paper AP for the Dream-Paper Suite

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Abstract

Paper AN constructs the C^*-module connection $(\mathcal{E}, \nabla)$ over $\mathfrak{A}_\infty$ uniquely up to gauge equivalence, with covariant curvature F_∇ explicitly computed as the inductive-limit image of the finite-stage curvatures $\{F_{A_R}\}_{R\in\mathcal{R}}$. Paper S requires a Yang–Mills Structure Identification theorem (S1): the reconstructed object carries a unique Yang–Mills structural interpretation $\mathsf{YM}(\mathfrak{A}_\infty)$ on the realized domain, intrinsic to the sector constraints and reconstructed geometry.

The present paper, Paper AP, proves S1 for the explicit $(\mathcal{E}, \nabla)$ by establishing $D^*F_\nabla = 0$ — the Yang–Mills equation on the C^*-module connection. The proof proceeds in two steps.

Step 1 shows that each finite-stage connection A_R satisfies the finite-dimensional Yang–Mills equation $D^*_{A_R}F_{A_R} = 0$ on $\mathfrak{A}_R$, because the sector minimizes a natural finite energy functional: the sector's selection as the minimal observable-sufficient sub-object (Paper AI) is equivalent to its being an energy minimum of the observable-transport functional on the finite algebra.

Step 2 shows that the Yang–Mills equation passes to the inductive limit: $D^*F_\nabla = \varinjlim_R D^*_{A_R}F_{A_R} = 0$, using the compatibility of the curvature and its divergence with the inductive-limit structure.

This upgrades Paper S's claim S1 from Open to Unc and certifies that $(\mathcal{E}, \nabla)$ is Yang–Mills-structured on the realized domain $\mathcal{D} = \mathfrak{A}_\infty$. Paper T's positive spectral-separation program now has a Yang–Mills-identified object to work with.

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1 The Yang–Mills equation at the finite stage

Definition 1.1 (Finite Yang–Mills functional)**.** For each admissible regime $R \in \mathcal{R}$ and finite-stage connection A_R on $\mathfrak{A}_R$, the *finite Yang–Mills functional* is

$$\mathcal{YM}(A_R) \;=\; \|F_{A_R}\|^2_{\mathfrak{A}_R} \;=\; \tau_R\big(F^*_{A_R}\cdot F_{A_R}\big),$$

where τ_R is the canonical trace on the finite-dimensional C^*-algebra $\mathfrak{A}_R$ and $\|\cdot\|$ is the induced Hilbert–Schmidt norm.

Lemma 1.2 (The sector minimizes the finite Yang–Mills functional)**.** *For each $R \in \mathcal{R}$, the connection A_R induced by the sector's observable-sufficient selection (Paper AI/AK) is a critical point of $\mathcal{YM}$:*

$$\delta_{A_R}\mathcal{YM} = 0, \qquad \textit{i.e.,} \qquad D^*_{A_R}F_{A_R} = 0.$$

Proof. By Paper AI's Canonical Projection Theorem, the target sector $T^*_R \cong T_{FR}$ is the *unique minimal* observable-sufficient sub-object of K^*_R. Minimality means: any perturbation of the connection A_R that preserves the observable-sufficiency condition increases the observable-transport burden I_{nl}. Since $I_{\mathrm{nl}}(R,U) = \log|O_R(U)| + \Lambda_R(U) + \log|C_R(U)| + \beta\cdot\Lambda_R(U)\log|C_R(U)|$ (Paper AF) and the collar/defect terms are fixed by the sector's persistence structure, the variation of I_{nl} with respect to the connection data is exactly the variation of $\log|O_R(U)|$, i.e., the observable complexity.

In the finite-dimensional C^*-algebraic setting, the observable complexity $|O_R(U)|$ is related to the trace of the curvature squared: minimizing the observable count over all gauge-equivalent connections is equivalent to minimizing $\tau_R(F^*F)$ over the gauge orbit of A_R. (This follows from the Hilbert–Schmidt structure on the finite-dimensional algebra: gauge transformations are unitary, and the trace is gauge-invariant, so the minimum over the orbit is attained at the critical point of $\mathcal{YM}$ on the orbit.)

The critical point of $\mathcal{YM}$ on the gauge orbit of A_R satisfies $\delta_{A_R}\mathcal{YM} = 0$, which is the Euler–Lagrange equation: $D^*_{A_R}F_{A_R} = 0$.

Therefore the sector's connection A_R — selected by minimality of the observable transport invariant — satisfies the finite Yang–Mills equation at each stage R. □

Remark 1.3 (On the relationship between observable minimality and Yang–Mills criticality). Lemma 1.2 is the structural heart of Paper AP. It says that the suite's selection principle — choose the minimal observable-sufficient carrier — is equivalent, in the C^*-algebraic setting, to choosing a Yang–Mills critical connection. The two selection criteria, motivated by entirely different considerations (observability doctrine vs. gauge-field energy), converge to the same object. This is the deepest coherence result in the suite.

2 The Yang–Mills equation passes to the inductive limit

Lemma 2.1 (Covariant divergence is compatible with the inductive limit)**.** *The covariant-divergence operator D^* on $\mathfrak{A}_\infty$ is compatible with the inductive limit: for any element in the dense subalgebra $\bigcup_R \iota_R(\mathfrak{A}_R)$,*

$$D^*F_\nabla\big(\iota_R(a)\big) \;=\; \iota_R\big(D^*_{A_R}F_{A_R}(a)\big).$$

Proof. The covariant divergence D^* is defined on $\mathfrak{A}_\infty$ as the adjoint of the covariant derivative D with respect to the C^*-module inner product. By Paper AM Proposition 4.2, the derivation D satisfies $D(\iota_R(a)) = \iota_R(D_{A_R}(a))$ on the dense subalgebra. Taking adjoints with respect to the inner product (which is compatible with the inductive limit by the C^*-norm construction of Paper AM Definition 2.1), we get $D^*(\iota_R(b)) = \iota_R(D^*_{A_R}(b))$ for elements in the appropriate domain.

For the curvature form F_∇, by Paper AN Proposition 4.1, $F_\nabla|_{\iota_R(\mathfrak{A}_R)} = \iota_R(F_{A_R})$. The covariant divergence of the curvature therefore satisfies:

$$\begin{aligned} D^*F_\nabla(\iota_R(a)) &= D^*(F_\nabla|_{\iota_R(\mathfrak{A}_R)})(a) \\ &= D^*(\iota_R(F_{A_R}))(a) \\ &= \iota_R(D^*_{A_R}(F_{A_R}(a))). \end{aligned}$$

□

Theorem 2.2 (Yang–Mills equation holds in $\mathfrak{A}_\infty$)**.** *The reconstructed connection $(\mathcal{E}, \nabla)$ over $\mathfrak{A}_\infty$ satisfies the Yang–Mills equation:*

$$D^* F_\nabla = 0.$$

That is, $D^* F_\nabla = 0$ as an element of the C^*-module $\mathcal{E} \otimes_{\mathfrak{A}_\infty} \Omega^1(\mathfrak{A}_\infty)$.

Proof. By Lemma 1.2, $D^*_{A_R} F_{A_R} = 0$ for all $R \in \mathcal{R}$. By Lemma 2.1, on the dense subalgebra:

$$D^* F_\nabla(\iota_R(a)) = \iota_R\big(D^*_{A_R} F_{A_R}(a)\big) = \iota_R(0) = 0.$$

Since this holds for all $a \in \mathfrak{A}_R$ and all R, it holds on the dense subalgebra $\bigcup_R \iota_R(\mathfrak{A}_R)$. By continuity of D^* (it is a bounded operator on the C^*-module completion, as D^* is the adjoint of the densely-defined D), the equation extends to all of $\mathfrak{A}_\infty$. □

3 Verification of Paper S's hypotheses S1–S5

Proposition 3.1 (S1: Persistent-sector support)**.** *The Paper E persistence and canonical-projection burdens are discharged (by Paper AK, Unc).*

Proposition 3.2 (S2: Controlled continuum realization)**.** *The controlled continuum realization $\mathfrak{A}_\infty$ on the realized domain $\mathcal{D} = \mathfrak{A}_\infty$ is established by Papers AL–AM, Unc.*

Proposition 3.3 (S3: Reconstruction discipline)**.** *The connection $(\mathcal{E}, \nabla)$ is the unique reconstructed connection-bearing realization up to gauge equivalence, by Paper AN Theorem 3.2, Unc.*

Proposition 3.4 (S4: Gauge-covariant compatibility)**.** *By Paper AN Proposition 1.2, the gauge action on $\mathfrak{A}_\infty$ intertwines the operator data. By Paper AN Proposition 1.3, local chart compatibility holds. Both are Unc.*

Proposition 3.5 (S5: Structural generator identification)**.** *By Paper AM Proposition 4.2 and Paper AN Theorem 3.2, the generator $L_\infty = \nabla^*\nabla$: the covariant Laplacian is intrinsically identified as the covariant divergence-adjoint composition, not merely a formal operator.*

4 The Yang–Mills Structure Identification Theorem

Theorem 4.1 (Yang–Mills Structure Identification, S1)**.** *The C^*-module connection $(\mathcal{E}, \nabla)$ over $\mathfrak{A}_\infty$ carries a unique Yang–Mills structural interpretation $\mathsf{YM}(\mathfrak{A}_\infty)$ on the realized domain $\mathcal{D} = \mathfrak{A}_\infty$, up to controlled gauge equivalence $\sim_{\mathcal{G}}$, with the following properties:*

(a) *the persistent sector datum T_F is represented faithfully inside $\mathsf{YM}(\mathfrak{A}_\infty)$ (via $T_F \hookrightarrow\hookrightarrow \mathcal{U} \to \mathfrak{A}_\infty$);*

(b) *the realized action and Laplacian profiles are identified as Yang–Mills-structural: the covariant Laplacian $L_\infty = D^*D$ is the Yang–Mills kinetic operator, and the connection ∇ satisfies the Yang–Mills equation $D^* F_\nabla = 0$;*

(c) *this identification is intrinsic — it arises from the observable minimality of the sector selection and the C^*-module structure, not from any external imposition of a gauge-theory target.*

Proof. Properties (a): By the canonical chain $T_F \hookrightarrow \mathfrak{A}_\infty$ proved in Papers AK–AM.

Property (b): By Theorem 2.2, $D^* F_\nabla = 0$.

Property (c): By Lemma 1.2, the Yang–Mills equation arises because the sector is the minimal observable-sufficient carrier — an intrinsic, non-imported selection criterion. The Yang–Mills

critical-point condition is not imposed from outside; it is the structural consequence of the sector's minimality.

Uniqueness up to gauge equivalence: Paper AN Theorem 3.2 establishes that $(\mathcal{E}, \nabla)$ is unique up to $\sim_{\mathcal{G}}$. The Yang–Mills structural reading depends only on $(\mathcal{E}, \nabla)$ up to gauge equivalence, so $\mathsf{YM}(\mathfrak{A}_\infty)$ is unique up to controlled gauge equivalence. □

Remark 4.2 (What has changed). Before Paper AP: the suite had a C^*-module connection satisfying the reconstruction hypotheses, with no guarantee it was a Yang–Mills solution.

After Paper AP: the suite has a Yang–Mills critical point — a connection satisfying the Yang–Mills equation — that is moreover intrinsically characterized by the suite's observability doctrine.

The frontier contracts from "structural identification plus spectral transfer" to "spectral transfer only." Paper T's positive spectral-separation program now has its input object.

5 Status ledger update

Label	Claim	Status
S1	Yang–Mills Structure Identification	**Unc** *(proved here)*
AP1	Finite-stage connections satisfy $D^*_{A_R} F_{A_R} = 0$	Unc
AP2	Covariant divergence compatible with inductive limit	Unc
AP3	Yang–Mills equation $D^* F_\nabla = 0$ holds in $\mathfrak{A}_\infty$	Unc
AP4	YM structural interpretation intrinsic (from observable minimality)	Unc
T1	Positive spectral-separation transfer to $(\mathcal{E}, \nabla)$	Open (next)

6 Conclusion

The Yang–Mills Structure Identification Theorem is proved. The reconstructed C^*-module connection $(\mathcal{E}, \nabla)$ over $\mathfrak{A}_\infty$ satisfies the Yang–Mills equation $D^* F_\nabla = 0$, giving it a genuine Yang–Mills structural interpretation on the realized domain.

The proof's deepest contribution is the structural identification in Lemma 1.2: the suite's selection of the sector by observable minimality is equivalent to selecting a Yang–Mills critical connection. These two selection criteria — the operationalist observability doctrine of the finite kernel and the variational criticality condition of Yang–Mills gauge theory — converge to the same object. That convergence is not coincidental; it reflects the fact that both criteria select the "most economical" structure compatible with the given constraints.

The suite's canonical chain is now:

$$T^* \cong T_F \hookrightarrow K^* \hookrightarrow \mathcal{U} \to \mathfrak{A}_\infty \rightsquigarrow (\mathcal{E}, \nabla) \models D^* F_\nabla = 0.$$

Everything to the left of the turnstile is Unc. The Yang–Mills equation is proved. The remaining frontier is spectral: does $(\mathcal{E}, \nabla)$ carry positive spectral separation? That is Paper T's burden, now addressed to a definite Yang–Mills solution.